

TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

ACADEMIC YEAR 2024/2025

MAY/JUNE EXAMINATION

AMIS1012 ETHICS IN COMPUTING

MONDAY, 26 MAY 2025

TIME: 6.30 PM – 8.30 PM (2 HOURS)

DIPLOMA IN COMPUTER SCIENCE

DIPLOMA IN INFORMATION TECHNOLOGY

Instructions to Candidates:

Answer **ALL** questions. All questions carry equal marks.

AMIS1012 ETHICS IN COMPUTING**Question 1**

- a) Discuss the differences among morals, ethics and law. Provide an example related to information technology for each term discussed. (9 marks)
- b) Discuss the following security attacks, and provide an example for each attack:
 - (i) Passive attacks (3 marks)
 - (ii) Active attacks (3 marks)
- c) List and describe **FIVE (5)** types of spams. (10 marks)

[Total: 25 marks]

Question 2

- a) International Organization for Standardization (ISO) is an international non-governmental organisation made up of national standards bodies. It is also one of the well-known industry standard security frameworks to provide a standardised approach related to information security management within companies.
Briefly explain ISO 27001 and list down any **THREE (3)** categories of ISO 27001. (5 marks)
- b) Briefly explain the **FOUR (4)** pillars of trustworthy computing. (8 marks)
- c) Discuss any **THREE (3)** IT security policies adopted by companies and provide an example for each to demonstrate its application in a company. (12 marks)

[Total: 25 marks]

Question 3

Ethan is a software development team leader. Ethan and his team are assigned by ABC Bank to develop a real-time Queue Management System, including a mobile application. Ethan and his team are known for their expertise in software development and software testing. The requirement of the system is well defined and the team has to complete their task within a year.

- a) Identify the suitable system development model for Ethan and his team. Provide relevant justification in your answer. (5 marks)
- b) Ethan and his team decided to use dynamic testing on the system they are developing. Explain the following types of dynamic testing and provide a scenario on how Ethan and his team can carry out each testing.
 - (i) Black-box testing (4 marks)
 - (ii) White-box testing (4 marks)

AMIS1012 ETHICS IN COMPUTING**Question 3 (Continued)**

- c) Describe **FOUR (4)** characteristics of a high-quality software product that Ethan and his team's system should exhibit. (12 marks)

[Total: 25 marks]

Question 4

Nintendo Co., Ltd. is a Japanese multinational video game company headquartered in Kyoto. It develops, publishes and releases both video games and video game consoles (such as Nintendo Switch). Its brand and products are protected by intellectual property law, which includes the name Nintendo, logos, characters design (eg. Pokemon, Super Mario, Zelda, and etc.), games and other marketing content. Individual and organisations are not entitled to use any of Nintendo's property without specific permission.

Source: Nintendo. (n.d.). *Nintendo intellectual property*. Nintendo.
<https://www.nintendo.com/au/legal/nintendo-intellectual-property/>

- a) Examine why Intellectual Property Rights (IPR) are important to Nintendo. Provide relevant example in your answer. (5 marks)
- b) Suggest **TWO (2)** most suitable types of IPR that Nintendo should consider for their market content. Provide relevant justification for your suggestion. (10 marks)
- c) Briefly explain **FIVE (5)** issues related to intellectual property protection. (10 marks)

[Total: 25 marks]